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Abstract

Blah blah blah

Definition. A completion of a metric space (X, d) consists of:

- (1) A complete metric space (Z, ρ) ;
- (2) An isometry $\iota : X \hookrightarrow Z$ such that $\overline{\iota(X)}^\rho = Z$.

We denote the completion of (X, d) by the ordered triple (Z, ρ, ι) .

Lemma. If (X, d) is a metric space, (Y, ρ) is a complete metric space, and $\iota : X \rightarrow Y$ is an isometry, then $(\overline{\iota(X)}^\rho, \rho, \iota)$ is a completion of (X, d) .

Theorem. Every metric space admits a unique completion up to isometric isomorphism.

Sketch of Proof. We will first show uniqueness, then existence.

Let (Y, ρ, ι) and (Z, σ, j) be completions of (X, d) .

Let $y \in Y$. Since $\iota(X)$ is dense in Y , there exists a sequence $\{\iota(x_n)\}_n \subset \iota(X)$ such that $\iota(x_n) \rightarrow y$.

Since the sequence $\{\iota(x_n)\}_n$ is convergent, it is ρ -Cauchy.

Since ρ and σ are isometries, we have:

$$\sigma(j(x_n), j(x_m)) = d(x_n, x_m) = \rho(\iota(x_n), \iota(x_m)).$$

Hence if $\{\iota(x_n)\}_n$ is ρ -Cauchy, then $\{j(x_n)\}_n$ is σ -Cauchy.

Since (Z, σ, j) is complete, $\{j(x_n)\}_n$ is convergent.

Define $\varphi : Y \rightarrow Z$ by $\varphi(y) = \lim_{n \rightarrow \infty} j(x_n)$. We have the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{\iota} & Y \\ & \searrow j & \vdots \varphi \\ & & Z \end{array}$$

To show uniqueness, show that φ is well-defined, an isometry, and bijective.

To show existence: Fix an $x_0 \in X$ and let $x \in X$ be arbitrary. Define $f_x : X \rightarrow \mathbb{R}$ by $f_x(t) = d(t, x) - d(t, x_0)$. Show that $f_x \in C_b(X)$ and that $\iota : X \rightarrow C_b(X)$ given by $x \mapsto f_x$ is an isometry. ■

This is called a Kuratowski embedding. Note that ι is not linear, hence if X was a normed vector space then its image in $C_b(X)$ would not necessarily be a vector space.

Question. Is the completion of a normed vector space also a vector space?

Answer. Yes! Though if we use a Kuratowski embedding on a normed vector space V , since ι is not linear the image of V may not necessarily be a vector space. We need to find a natural linear isometry into a Banach space.

Definition. Let $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$ be normed vector spaces.

- (1) The operator norm of $T \in \text{Hom}(V, W)$ is $\|T\|_{\text{op}} := \sup\{\|T(v)\| \mid v \in V, \|v\| \leq 1\}$.
- (2) $T \in \text{Hom}(V, W)$ is called bounded if $\|T\|_{\text{op}} < \infty$. The set of all bounded linear maps from V to W is $\mathcal{B}(V, W) := \{T \in \text{Hom}(V, W) \mid \|T\|_{\text{op}} < \infty\}$.
- (3) The continuous dual of V is $V^* := \mathcal{B}(V, \mathbb{F})$.

Remarks.

- (1) A linear map is continuous if and only if it is bounded.
- (2) $(\mathcal{B}(V, W), \|\cdot\|_{\text{op}})$ is a Banach space.
- (3) Is V^* nonempty?

Theorem (Hahn-Banach). Let X be an $\mathbb{R}$ -vector space, M a subspace of X , and $p : X \rightarrow \mathbb{R}$ a seminorm on X . If $f : M \rightarrow \mathbb{R}$ is a linear functional on M such that $|f(m)| \leq p(m)$ for all $m \in M$, then there exists a linear functional $F : X \rightarrow \mathbb{R}$ such that $F|_M = f$ and $|F(x)| \leq p(x)$ for all $x \in X$.

The assumption that $|f(m)| \leq p(m)$ implies f is bounded. The claim that $|F(x)| \leq p(x)$ implies F is bounded.

Corollary. Let $(V, \|\cdot\|)$ be a normed vector space. For any nonzero $v \in V$, there exists $F \in V^*$ such that $\|F\|_{\text{op}} = 1$ and $F(v) = \|v\|$.

Sketch of Proof. Fix $v \in V$. Define $M := \{\alpha v \mid \alpha \in \mathbb{R}\}$. This is a subspace of V .

Define $f : M \rightarrow \mathbb{R}$ by $f(\alpha v) := \alpha \|v\|$. This map is linear.

Note that $|f(\alpha v)| = |\alpha| \|v\| = \|\alpha v\|$. Since every norm is a seminorm, the hypothesis of the Hahn-Banach theorem is satisfied.

Observe that:

$$\begin{aligned} \|f\|_{\text{op}} &= \sup \left\{ |f(\alpha v)| \mid \alpha v \in M, \|\alpha v\| \leq 1 \right\} \\ &= \sup \left\{ \|\alpha v\| \mid \alpha v \in M, \|\alpha v\| \leq 1 \right\} \\ &= 1. \end{aligned}$$

Take $\alpha = 1$, then $f(v) = \|v\|$.

By the Hahn-Banach Theorem, f extends to a linear functional $F \in V^*$. ■

Corollary. Let $(V, \|\cdot\|)$ be a normed $\mathbb{R}$ -vector space and $v \in V$. Then $\|v\| = \sup\{|\varphi(v)| \mid \varphi \in V^*, \|\varphi\|_{\text{op}} \leq 1\}$.

Proof. If $\varphi \in V^*$ and $\|\varphi\| \leq 1$, then $|\varphi(v)| \leq \|\varphi\|_{\text{op}} \|v\| \leq \|v\|$. By the previous corollary, □